

Object Detectors for Side-Scan Sonar Mine Detection

Aditya Koushik¹ Elvin Fu² Max Weinacht³

¹Computer Science ²Electrical Engineering ³Physics, Stanford University
{akoushik, elvinf, maxw2}@stanford.edu



Background

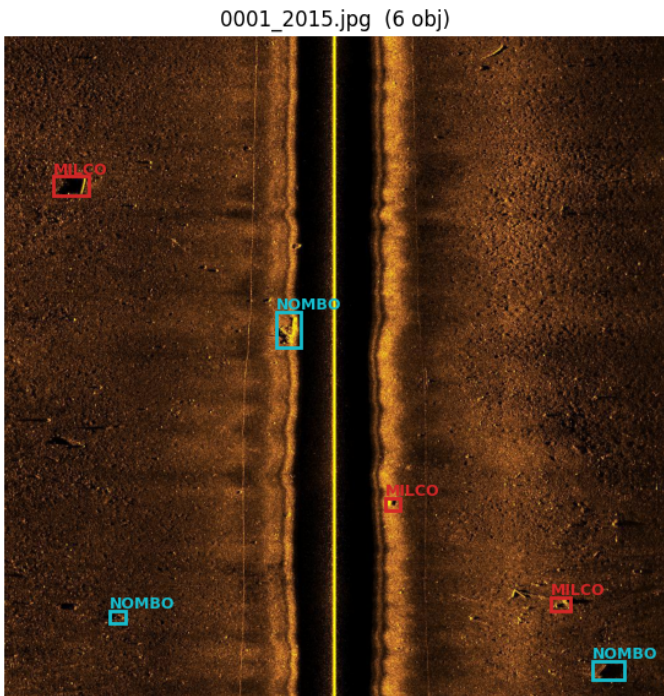
Even after conflicts end, sea mines remain a long-lived risk for shipping, fishing, and coastal infrastructure. We built a whole-image object detector for side-scan sonar (SSS) that takes 800×800 sonar tiles as input and outputs MILCO/NOMBO (mine/not-mine) bounding boxes with confidence scores. The problem is hard because labeled sonar data are scarce and SSS texture includes speckle, range-dependent intensity, highlight–shadow geometry, and seabed clutter unlike natural images. Holding YOLOv8n fixed, sonar-aware augmentation plus supervised Cylinder→Santos transfer reached .761 held-out $mAP_{0.5}$, compared with .539 for no augmentation. We ask whether augmentation closes the natural-image-to-sonar gap and which pretrained weights transfer best.

Data and Features

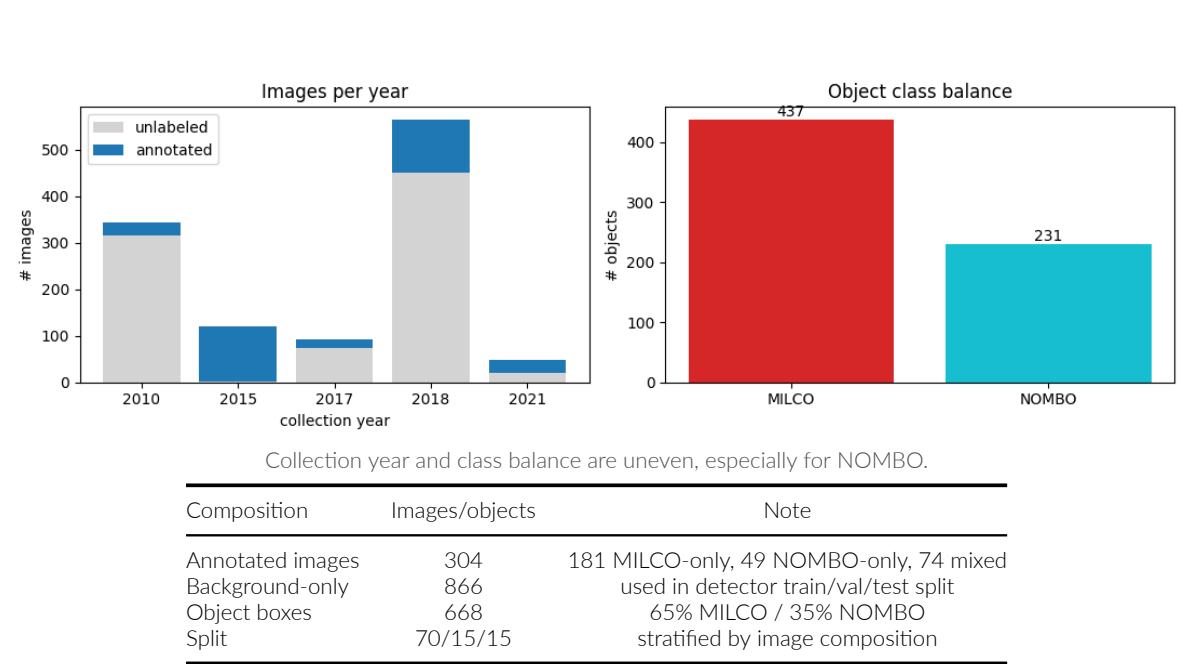
Data: we focus primarily on the Santos SSS benchmark, collected by the Portuguese Navy between 2010 and 2021. The benchmark includes MILCO (mine-like contact), NOMBO (non-mine bottom object), and many background images, making it small, imbalanced, and operationally realistic.

- Detection target:** predict a bounding box and classify each object as MILCO or NOMBO.
- Santos benchmark:** 1,170 images; 304 annotated; 668 boxes split into 437 MILCO and 231 NOMBO.
- Auxiliary sonar data:** BenthicCat supplies unlabeled SSS texture for BYOL pretraining; Cylinder2 supplies supervised small-object sonar detections before Santos fine-tuning.

Features: the raw inputs are normalized SSS image tiles, while derived supervision consists of object boxes, MILCO/NOMBO labels, and detector feature maps learned during training. SSS imagery differs sharply from natural images: coherent acoustic scattering creates multiplicative speckle, sonar range changes brightness, targets appear as highlight–shadow structures, and seabed textures have no clean ImageNet analog. Cylinder2 contributes 478 side-scan images across cylinder and manta-ray classes, each containing one small object; BenthicCat contributes large-scale unlabeled SSS tiles for self-supervised texture learning, but no mine targets.



Example labeled scan: MILCO in red, NOMBO in cyan.



Related Work

Santos et al. introduced the SSS benchmark and reported an initial YOLOv4 baseline, but the result was preliminary and not tied to a clean held-out protocol. Later work on the same data has largely improved the detector architecture itself: SOCA-YOLO combines restoration, SPDConv, and CBAM attention; YOLOv11-SDC adds sparse feature extraction, dilated reparameterized blocks, and content-guided fusion; ESL-YOLO adds Sobel edge fusion, self-calibrated dual attention and a location-quality estimator.

Mine-JEPA studies the related patch-classification problem with in-domain self-supervised learning, but that setting assumes the objects are already localized. Our project keeps the detector architecture fixed and asks a different question: how much of the sonar domain gap can be closed by data, augmentation, pretraining, and evaluation protocol alone? This framing makes the comparison useful even when stronger detectors are released in the future.

The major protocol gap in prior comparisons is that reported metrics often mix validation-style reporting, cross-dataset testing, and single-split numbers on a heterogeneous dataset. We therefore chose to separate architecture from data choices and report held-out, seed-averaged and fold-level uncertainty where it was possible.

Models and Metrics

- Detector:** YOLOv8n, roughly 3.2M parameters, 800×800 inputs, batch 16 and early stopping on validation mAP.
- Split:** stratified 70/15/15 train/val/test by background, MILCO-only, NOMBO-only and mixed-image strata; best model rechecked with stratified 5-fold evaluation.
- Reason for YOLOv8n:** Santos has only a few hundred annotated images, so a small convolutional detector is easier to optimize than a larger transformer-style detector.
- Box match:** prediction must have correct class and $IoU \geq \tau$ against an unmatched ground-truth box.

We used Ultralytics detection metrics. Predictions are matched in descending confidence order, and unmatched predictions are false positives, and unmatched ground truths are false negatives. Reported precision and recall are taken at the F1-maximizing confidence threshold.

$$IoU(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}, \quad P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}$$

$$AP_{\tau} = \int_0^1 P(R) dR, \quad mAP_{0.5:0.95} = \frac{1}{10} \sum_{\tau=0.5}^{0.95} mAP_{\tau}$$

YOLOv8 uses a convolutional backbone with C2f blocks and an SPPF module, followed by a PAN/FPN-style neck that carries both semantic and lower-level edge/detail information into the detection heads. This is useful for SSS because mine-like objects are small and often defined by sharp highlights, shadows nearby, or weak boundaries against other clutter on the sea bed.

Models: Training Objective and Protocol

Table 1. Experiment matrix from the writeup.

ID	Arch.	Backbone Init	Head Init	Augmentation
<i>Experiment 1: Augmentation ablation (fixed COCO YOLOv8n init)</i>				
A0	YOLOv8n	COCO	COCO	None
A1	YOLOv8n	COCO	COCO	Generic CV (flips, translate, scale, mosaic)
A2	YOLOv8n	COCO	COCO	Speckle noise
A3	YOLOv8n	COCO	COCO	Range-dependent falloff
A4	YOLOv8n	COCO	COCO	Acoustic shadow synthesis
A5	YOLOv8n	COCO	COCO	Physics stack (A2+A3+A4)
A6	YOLOv8n	COCO	COCO	Generic CV + physics stack
A7	YOLOv8n	COCO	COCO	Generic CV + speckle + range falloff
A8	YOLOv8n	COCO	COCO	All augmentations (A1+A2+A3+A4)
<i>Experiment 2: Pretrained-weight source (augmentation fixed to A*)</i>				
B1	YOLOv8n	Random	Random	A*
B2	YOLOv8n	COCO (backbone+neck)	Random	A*
B3	YOLOv8n	COCO (backbone+neck)	COCO ¹	A*
B4	YOLOv8n	Self-sup. BenthicCat SSS	Random	A*
B5	YOLOv8n	Cylinder/manta SSS (supervised)	Cyl./manta ¹	A*
B6	YOLO26n	COCO (backbone+neck)	COCO ¹	A*
B7	YOLO26n-P2	COCO partial transfer	Random	A*
<i>Experiment 3: Stratified k-fold cross-validation (k = 5)</i>				
C1	YOLOv8n	Best of Exp. 2	Best of Exp. 2	A* (5-fold stratified CV)
<i>Experiment 4: Classical baseline (classification probe)</i>				
D1	YOLOv8n (frozen)	COCO (layers 0–9)	–	256-d embedding → logistic regression

Methods - continued

Formulas. YOLOv8 classification uses BCE on target-assignment scores; boxes use CloU plus Distribution Focal Loss:

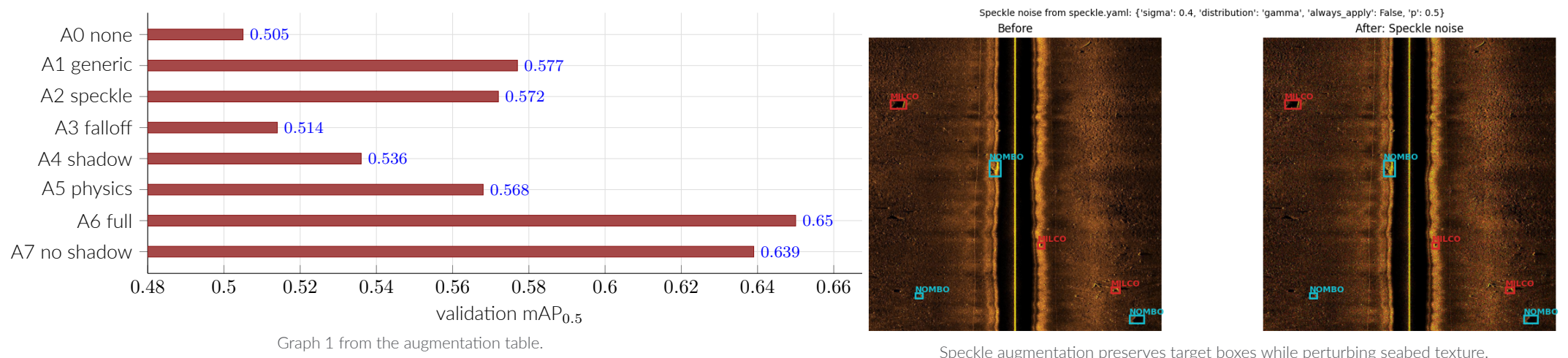
$$L_{cls} = \sum_i BCEWithLogits(s_i, s), \quad L_{box} = \sum_i (1 - \text{CloU}(b_i, b))w_i, \\ L_{obj} = CE(p_i, y_i)(y_i - y) + CE(p_i, y_i)(y - y_i), \\ L = \lambda_{box}L_{box} + \lambda_{cls}L_{cls} + \lambda_{obj}L_{obj}, \\ \xi \leftarrow \tau\xi + (1 - \tau)\theta, \\ L_{BYOL} = 2 - 2 \frac{qy(gy(Js(v_1))) \cdot qe(Je(v_2))}{\|qy(gy(Js(v_1)))\| \|qe(Je(v_2))\|} + (1 \leftrightarrow 2), \\ x = \text{flatten}(\text{AAP}(F_{obj}(I))) \in \mathbb{R}^{256}, \\ \min_w - \sum_i \alpha_i \log \frac{\exp(W_i^T x_i)}{\sum_j \exp(W_j^T x_i)} + \lambda \|W\|_2^2$$

Training protocol. We tuned only learning rate and layer freezing after choosing the augmentation stack. Full fine-tuning beat freezing the first ten layers at every learning rate tested, with $\text{lr}_0 = 0.01$ giving the best validation $mAP_{0.5}$. The epoch cap was made intentionally high to be early stopping, not on a fixed schedule, terminating the training.

The classification branch is reinitialized for the two Santos classes at fine-tuning time. For transfer settings with pretrained detector heads, box regression weights carry over when the shapes match, while the class branch is adapted to MILCO/NOMBO. This separates transferable localization from transferable class semantics, avoiding a kind of “contamination”.

Results 1: Augmentation Sweeps

We carried the best augmentation stack, A*, into all later detector experiments: generic flips/scale/translate/mosaic plus speckle, range falloff and synthetic acoustic shadows. Mosaic exposed the detector to varied object scales, seabed textures and image compositions; vertical flips were allowed because SSS scans have no natural-image notion of “upside-down”.



Interpretation: generic transforms provided the largest single gain; physics transforms were complementary and reduced overfitting to the limited Santos appearance distribution.

Generic CV was especially effective because the annotated set is small: flips, translation, scaling and mosaic force the detector to see targets in different positions, scales and seabed contexts. Physics-only augmentation improved precision and stricter localization, but it didn’t replace the geometric/compositional diversity from generic CV. The selected A* therefore used both families rather than treating sonar physics as a stand-alone substitute for standard detection augmentation.

A6 reached .650 validation $mAP_{0.5}$, a +.145 gain over no augmentation and +.073 over generic CV alone. Mosaic probability was then swept inside A6; mosaic 1.0 was selected because it tied the best $mAP_{0.5}$ while giving the best $mAP_{0.5:0.95}$ and lowest seed variance.

Stack	$mAP_{0.5}$	$mAP_{0.5:0.95}$
None	.505	.300
Generic	.577	.331
Physics	.568	.376
Full A*	.650	.377

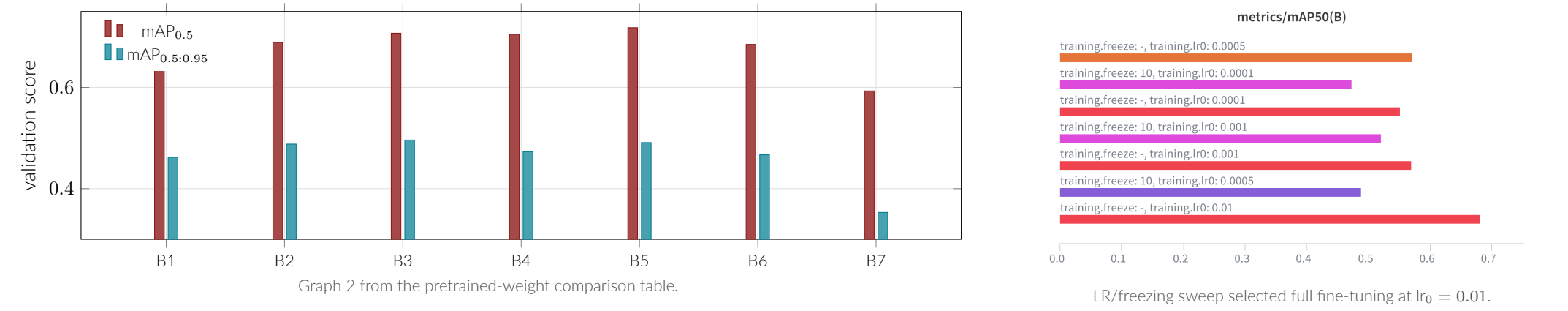
Mosaic prob.	$mAP_{0.5}$	$mAP_{0.5:0.95}$	P	R
0.25	.714 ± .075	.472 ± .022	.860 ± .001	.678 ± .055
0.50	.688 ± .002	.408 ± .031	.748 ± .003	.661 ± .002
1.00	.719 ± .046	.477 ± .006	.858 ± .010	.656 ± .034

Speckle was applied multiplicatively so the average brightness is preserved while seabed texture changes. Range falloff is applied along the sonar range axis to imitate attenuation or imperfect time-varied gain correction. Shadow synthesis darkens a soft strip behind annotated boxes away from the nadir line, exposing the detector to different acoustic-shadow strengths without changing target labels.

Transform	Writeup definition used in training
Speckle	multiplicative coherent-acoustic noise: $I' = I \cdot S$, with Gamma field normalized to unit mean.
Range falloff	range-axis attenuation, $I'(r) = I(r)(1 + r)^{-\alpha}$, approximating distance-dependent intensity rolloff.
Shadow	soft dark strip extending away from nadir behind each box; boxes unchanged.

Results 2: Pretrained-Weight Source

Experiment 2 isolated the source of initial weights while holding the augmentation fixed at A*. B1–B3 asks how much COCO transfer helps (none, backbone/neck only, or full detector); B4 tests sonar-texture BYOL; B5 tests supervised in-domain detection transfer; B6–B7 test newer YOLO26 variants without actually changing the Santos protocol.



Setting	$mAP_{.5}$	$mAP_{.5:.95}$	P	R
B1 rand.	.699 ± .049	.462 ± .050	.819 ± .030	.645 ± .081
B2 COCO tb.	.680 ± .056	.485 ± .060	.838 ± .111	.665 ± .072
B3 COCO det.	.707 ± .043	.496 ± .066	.842 ± .035	.675 ± .038
B4 BYOL	.705 ± .064	.473 ± .068	.832 ± .066	.629 ± .045
B5 Cyl→San.	.718 ± .034	.491 ± .028	.886 ± .018	.652 ± .016
B6 Y26	.685 ± .022	.467 ± .027	.822 ± .055	.665 ± .012
B7 Y26-P2	.698 ± .103	.480 ± .101	.706 ± .143	.671 ± .075

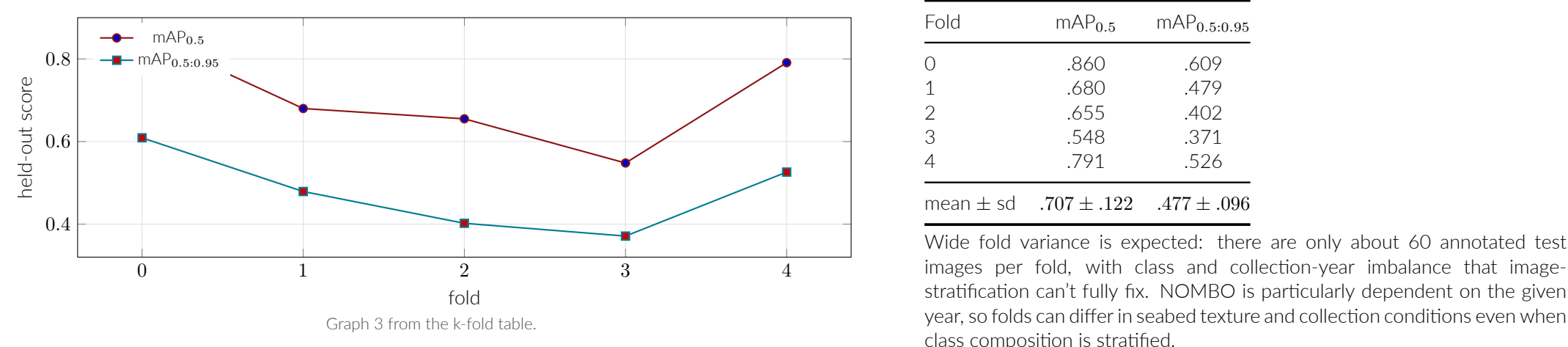
Why B5 wins: Cylinder2 is supervised sonar detection with small, high-contrast highlight–shadow targets, so it warms up objectness and box localization before the final Santos two-class fine-tune. BYOL on BenthicCat is sonar-domain data, but it sees mostly habitat texture and has no MILCO/NOMBO localization objective, so the random detector head still has to learn the hard part from just a few labels.

The other YOLOv8 variants were close enough that their ordering was not very meaningful without more seeds. Random initialization was surprisingly competitive, likely because YOLOv8n is small and the augmentation stack regularizes the limited data. COCO helped more on the stricter $mAP_{0.5:0.95}$ localization metric than on $mAP_{0.5}$, suggesting that natural-image pretraining transfers box geometry better than sonar object appearance. YOLO26n and YOLO26n-P2 did not help here; the P2 small-object head adds capacity that this dataset may not support.

On held-out test, B5 is best across $mAP_{0.5}$ for both MILCO and NOMBO, $mAP_{0.5}$, $mAP_{0.75}$, and $mAP_{0.5:0.95}$. Its precision advantage is especially important for mine-search workflows because false positives consume scarce review and disposal resources.

The BenthicCat result is interesting: sonar-domain pixels alone are not enough if the pretext task does not teach objectness. Conversely, Cylinder2 is much smaller but detection-shaped, so it provides a better warm start for localization and object/no-object decisions.

Results 3: Stratified k-Fold Evaluation



The 5-fold mean $mAP_{0.5}$ of .707 closely matched B5 validation (.718), so the fixed split was not simply lucky; the important lesson is uncertainty. Single-run Santos results can look decisive while fold $mAP_{0.5}$ ranges from .548 to .860.

This spread is a property of the dataset, not just the model. Each fold has only about 60 annotated test images, and stratifying by image composition cannot fully balance collection year, seabed texture or the fact that NOMBO examples are concentrated in a very small number of years.

Practically, this means future Santos papers should avoid relying on a single validation split or a single seed. Cross-year validation would be especially informative because it would explicitly test whether a detector has learned target evidence or year-specific sonar/seabed conditions.

Results 4: Training Curves and Logistic Baseline



For the logistic-regression probe, each ground-truth crop is embedded by frozen YOLOv8 backbone layers 0–9 and global-average-pooled, then a class-balanced multinomial logistic model predicts MILCO vs. NOMBO (shown in Methods). The probe is intentionally classification-only: it measures feature separability after the detector is handled perfect boxes. MILCO is easier than NOMBO in this probe, which is consistent with the class imbalance and lower NOMBO support. Test accuracy is 65.6%, with test AUROC 73.6%, showing a useful class signal in frozen COCO YOLO features but not enough to solve whole-image detection without objectness and localization.

The baseline also clarifies the difference between patch classification and mine detection: a classifier can exploit already-centered object crops, but an operational detector must find rare small objects in full scans and avoid fring on background-only images.

Discussion: Comparison to Prior Work

0.761

held-out test mAP_{0.5}
B5 Cyl→Santos

+0.222

gain over YOLOv8n A0
no augmentation

0.548

held-out mAP_{0.5:0.95}
strict localization

Method	Base	P	R	mAP _{0.5}	mAP _{0.5:0.95}
SOCA-YOLO	YOLOv9	93.7	76.2	83.7	–
YOLOv11-SDC	YOLOv11s	93.4	69.8	82.5	59.8
ESL-YOLO	YOLOv11s	87.7	75.6	84.7	58.4
YOLOv8n A0	no aug	63.7	51.5	53.9	35.0
Ours B5	data only	88.2	65.2	76.1	54.8

The published methods improved the architecture with sonar-specific modules such as edge fusion, specialized attention, restoration and quality estimation. Our result is lower than those headline $mAP_{0.5}$ values, but it changes only data and initialization while using a stricter held-out protocol and reporting seed/fold uncertainty. Our most meaningful gain is internal: holding YOLOv8n fixed, the system rises from .539 $mAP_{0.5}$ with no augmentation to .761 held-out $mAP_{0.5}$ with sonar-aware augmentation and Cylinder→Santos transfer.

Future Work / Conclusion

In conclusion, when holding architecture fixed choice of data limits the impact of moving from natural images to sonar images. The best detector is B5, the supervised Cylinder→Santos transfer model. It’s advantages likely come from warming up objectness and localization on a small-object sonar detection task before it gets its final Santos fine-tuning.

- Training data mattered more than architecture swaps:** YOLO26n did not beat YOLOv8n on this benchmark.
- In-domain supervised transfer beat in-domain BYOL:** detection-shaped sonar pretraining helped more than texture-only self-supervision.
- Report uncertainty, not only best split:** fold $mAP_{0.5}$ varied from .548 to .860.
- Next:** cross-year validation, more seeds, semi-supervised use of unlabeled Santos backgrounds and combining data adaptation with sonar-specific detector modules.

Ultimately, we upgraded from a weak no-augmentation baseline to a fairly-accurate model by using a simple detector combined with sonar-specific training. This suggests that future gains should combine stronger sonar architectures with careful data protocol rather than treating architecture alone as the main source of progress.

With more compute, the natural next step would be to combine supervised sonar transfer with unlabeled Santos backgrounds, pretraining objectives closer to detection (segmentation, box prediction or MILCO/NOMBO classification), and stronger backbones such as DINO-style vision transformers or RT-DETR-style detectors evaluated under the same stratified/cross-year protocol.

References

[1] Santos et al., *Data in Brief*, 2024. Side-scan sonar mine-detection dataset.
[2] Blondel, *The Handbook of Sidescan Sonar*, Springer, 2010.
[3] Steinger et al., *Engineering Applications of AI*, 2022. Deep learning for sonar imagery.
[4] Jocher et al., Ultralytics YOLOv8, GitHub, 2023.
[5] Grill et al., NeurIPS, 2020. Bootstrap Your Own Latent.
[6] Rajani et al., arXiv, 2025. BenthicCat opti-acoustic dataset.
[7] yeesonmin@naver.com, Roboflow Universe, 2022. Cylinder2 dataset.
[8] Cui et al., *Frontiers in Marine Science*, 2025. SOCA-YOLO.
[9] Zou et al., *Journal of Marine Science and Engineering*, 2025. YOLOv11-SDC.
[10] Zhang et al., *Journal of Marine Science and Engineering*, 2025. ESL-YOLO.
[11] Kwon et al., arXiv, 2026. Mine-JEPA patch classification.